Lecture # 08

MM 318: Multicomponent Alloys

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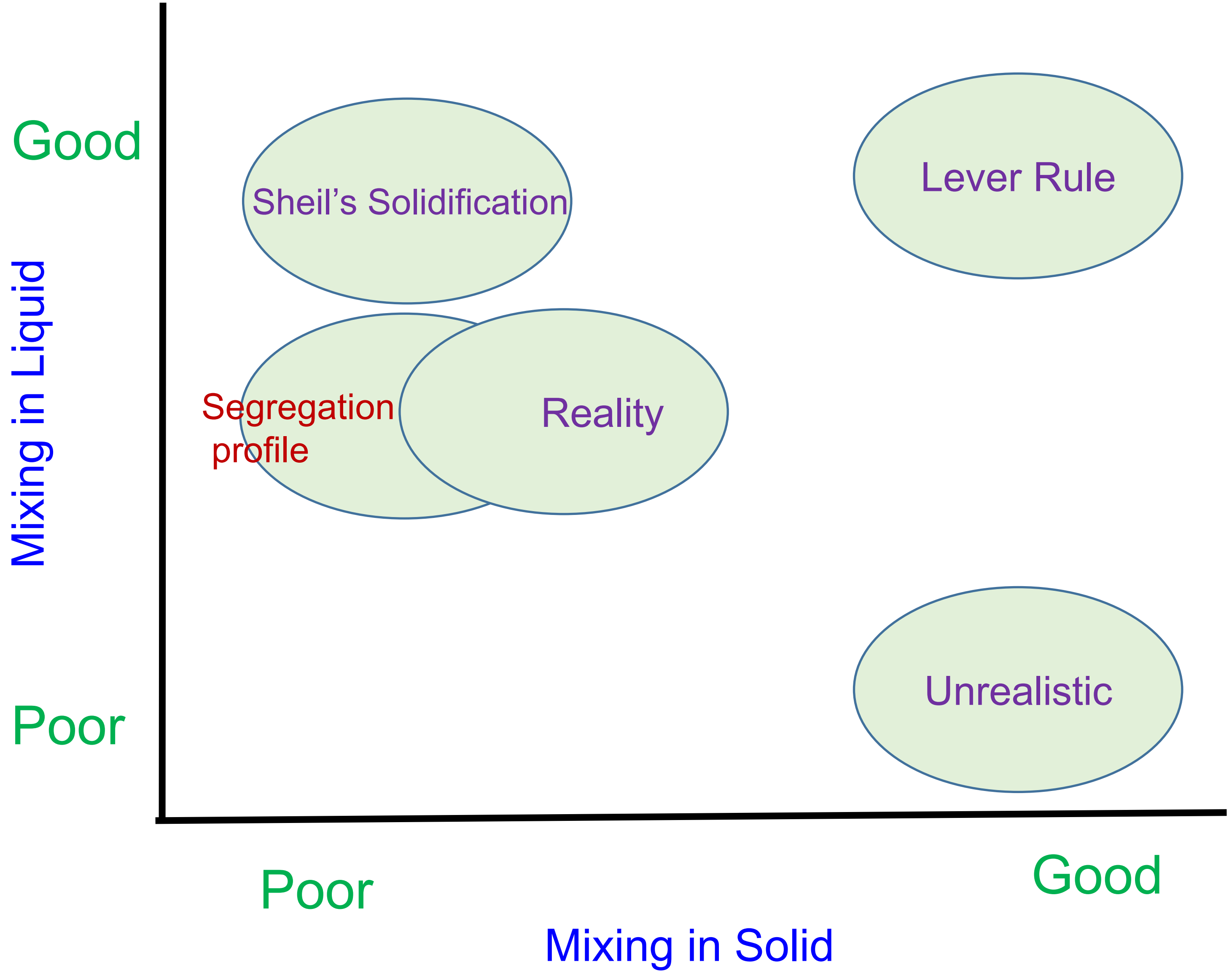
Microstructural evolution during solidification



Solidification at equilibrium condition

1. Lever Rule

(Complete diffusion in both Solid and Liquid)

$$f_s = \frac{C_L - C_o}{C_L - C_s}, \qquad f_L = \frac{C_o - C_s}{C_L - C_s}$$

Solidification at non-equilibrium condition

1. **Scheils' Equation**

(No diffusion in Solid, but complete diffusion in Liquid)

$$C_L = \frac{C_o}{(1 - f_s)^{1-k}}$$

2. **Segregation profile**

(No diffusion in Solid, but limited diffusion in Liquid)

$$\frac{C_L}{C_o} = [1 + \frac{1-k}{k} e^{-vx/D}]$$

Partition coefficient: $k = \frac{C_S}{C_L}$

Temperature gradient (G*)

$$G^* = \left(\frac{\partial T_L}{\partial x}\right)_{interface}$$

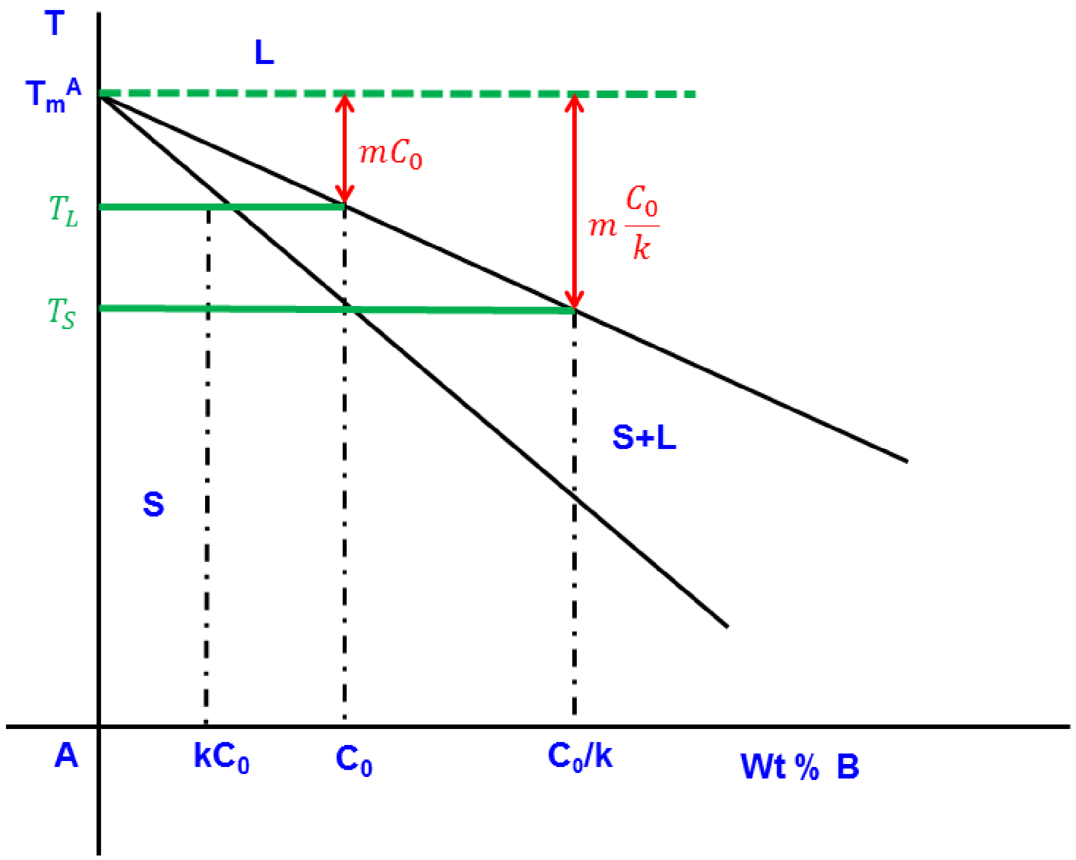
$$G^* = \frac{\partial T_L}{\partial C_L}\,\frac{\partial C_L}{\partial x}$$

$$G^* = -m\,\frac{\partial C_L}{\partial x}$$ (Where $m = \frac{\partial T_L}{\partial C_L}$; $Negative\ Slope$)

$$G^* = \frac{\partial}{\partial x}(-mC_L)$$

$$G^* = \frac{\partial}{\partial x}\left(-mC_0\left[1+\left(\frac{1-k}{k}\right)\exp\left(-\frac{vx}{D}\right)\right]\right)$$

$$G^* = -mC_0\left(\frac{1-k}{k}\right)\left(-\frac{v}{D}\right)\exp\left(-\frac{vx}{D}\right)$$



$$G^* = -mC_0 \left(\frac{1-k}{k}\right)\left(-\frac{v}{D}\right)\exp\left(-\frac{vx}{D}\right)$$

At the interface; x=0

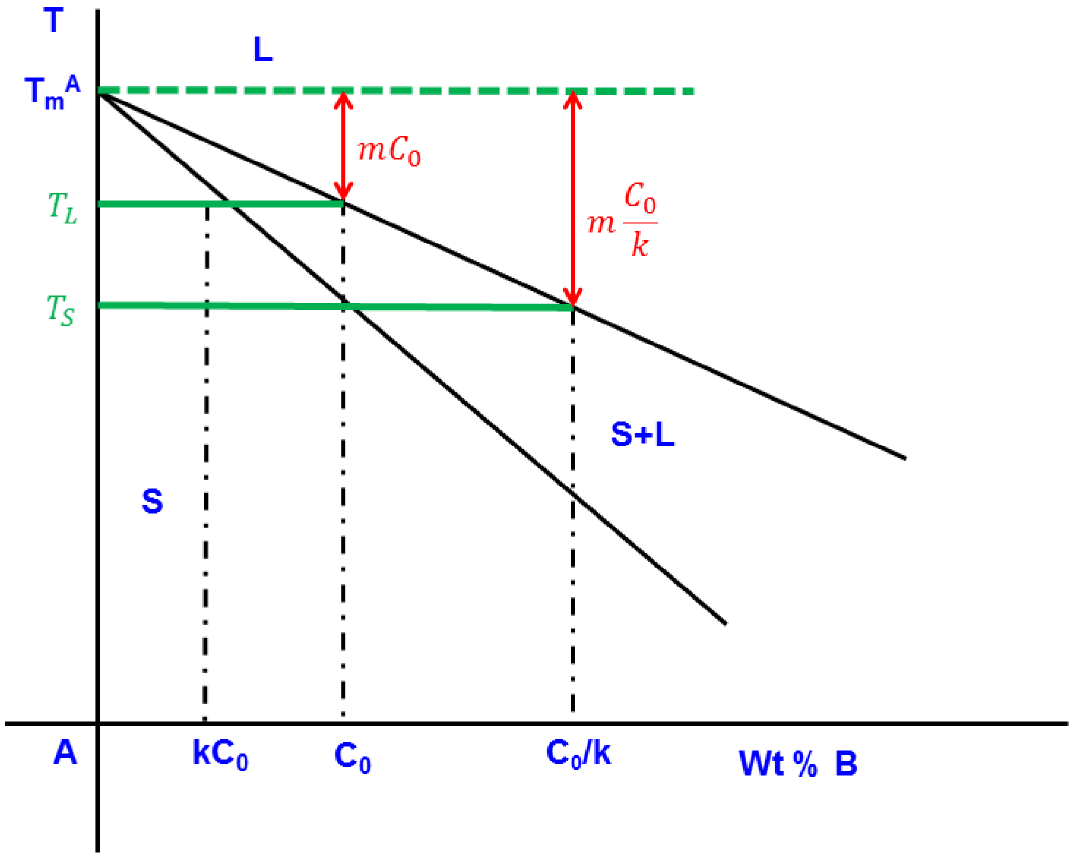
$$G^* = -mC_0 \left(\frac{k-1}{k}\right)\left(\frac{v}{D}\right)$$

$$G^* = -mC_0 \left(1-\frac{1}{k}\right)\left(\frac{v}{D}\right)$$

$$G^* = \left(-mC_0 - \left(\frac{-mC_0}{k}\right)\right)\left(\frac{v}{D}\right)$$

$$G^* = (T_L - T_S)\left(\frac{v}{D}\right)$$

$$\Rightarrow \frac{G^*}{v} = \frac{T_L - T_S}{D}$$



$$\frac{G^*}{v} = \frac{T_L - T_S}{D} = \frac{\Delta T}{D}$$

Where G^*= Temperature gradient, v = Velocity of interface, $T_L - T_S$: Freezing range, D : Diffusion coefficient

- $\frac{G}{v}$ determines the morphology of solidification microstructure

$(i)\ \ G\ >\ G^* : Plannar$

$(ii)\ G\ <\ G^* : Cellular/Dendrite$

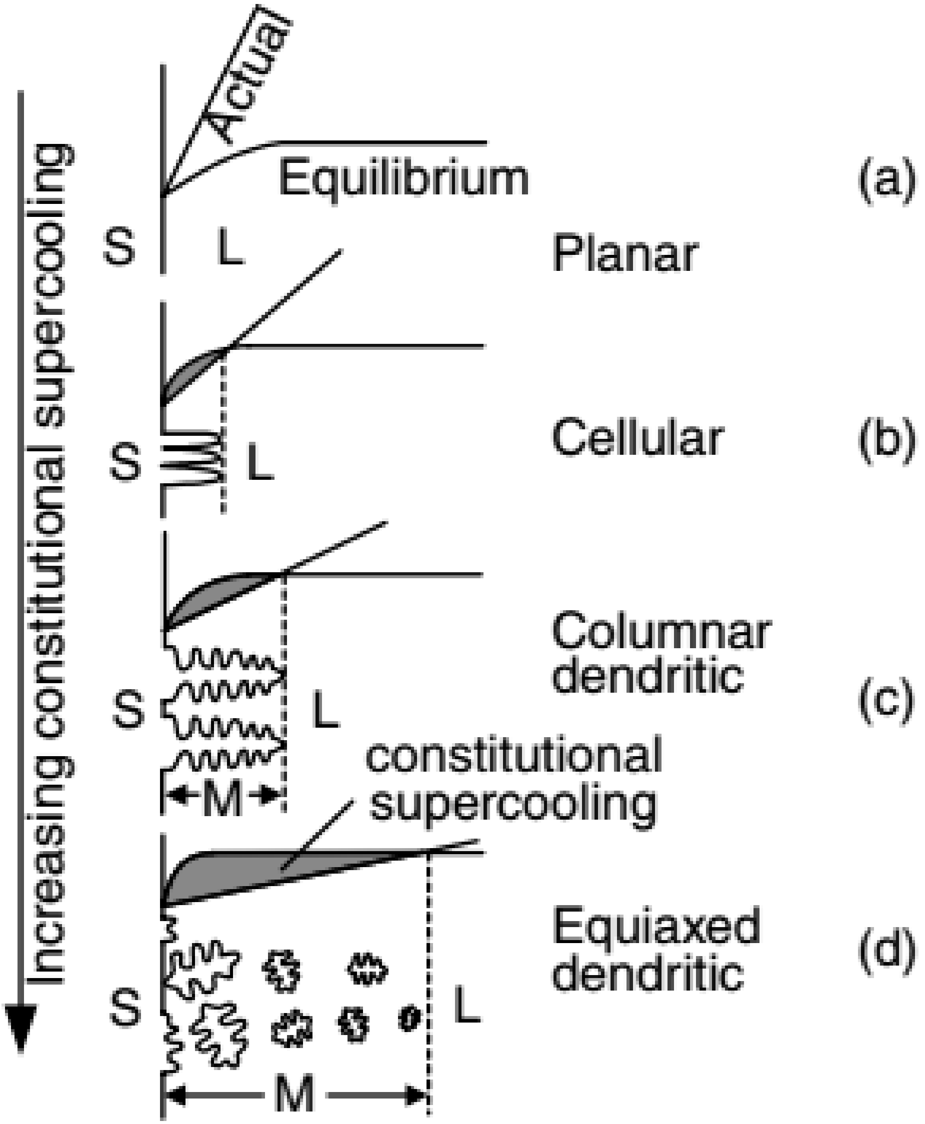
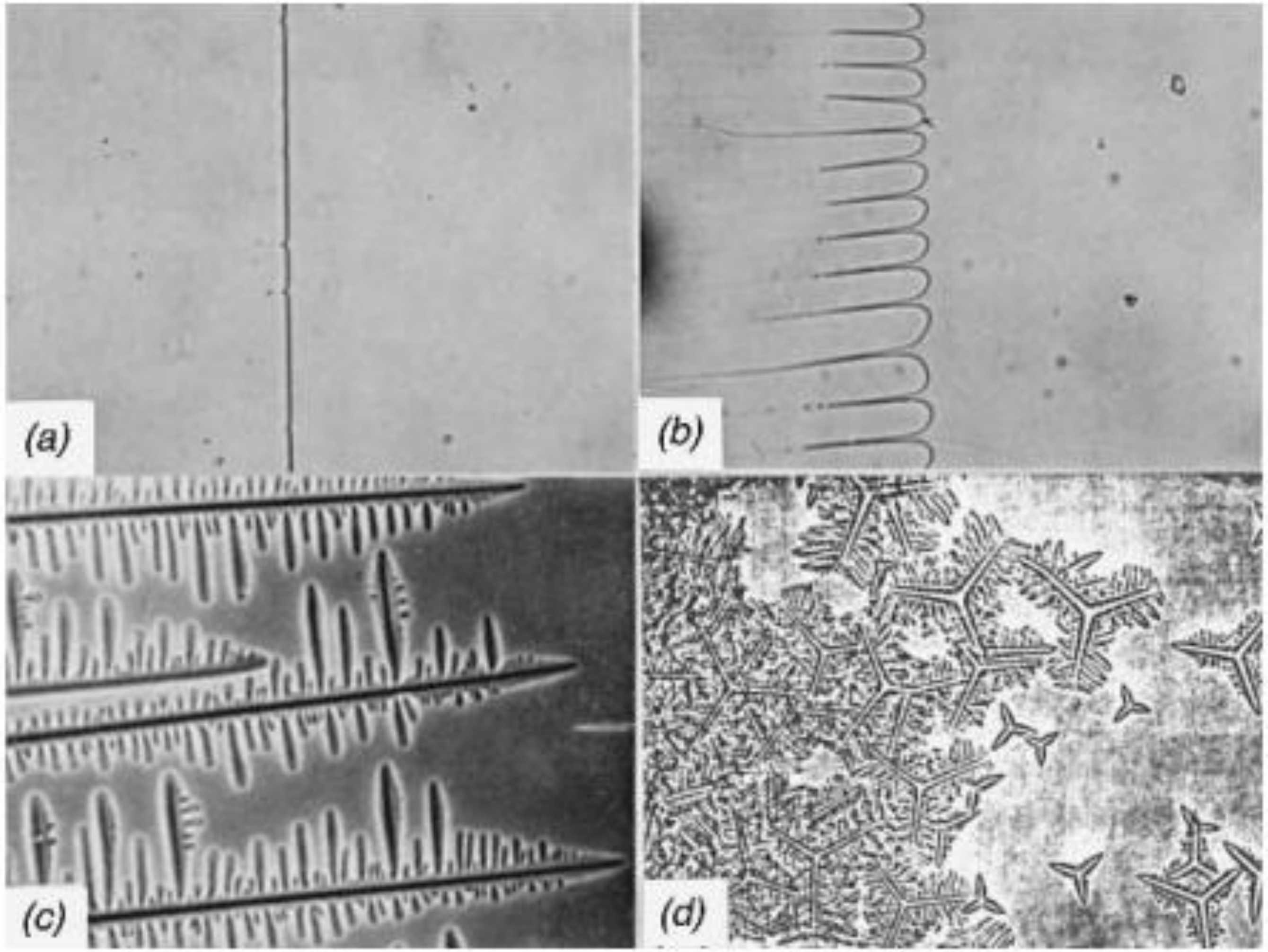


Figure: Effect of constitutional supercooling on solidification mode: (a) planar; (b) cellular; (c) columnar dendritic; (d) equiaxed dendritic (S, L, and M denote solid, liquid, and mushy zone, respectively.

***Reference**: Sindo Kou, Welding Metallurgy, Second Edition, 2003, John Wiley & Sons, Inc., ISBN: 0-471-43491-4*



Basic solidification modes (Magnification 67X):

(a) Planar solidification of carbon tetrabromide;
(b) Cellular solidification of carbon tetrabromide with small amount of impurity;
(c) Columnar dendritic solidification of carbon tetrabromide with several percent impurity;
(d) Equiaxed dendritic solidification of cyclohexanol with impurity.

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Cooling rate ($\dot{T}$)

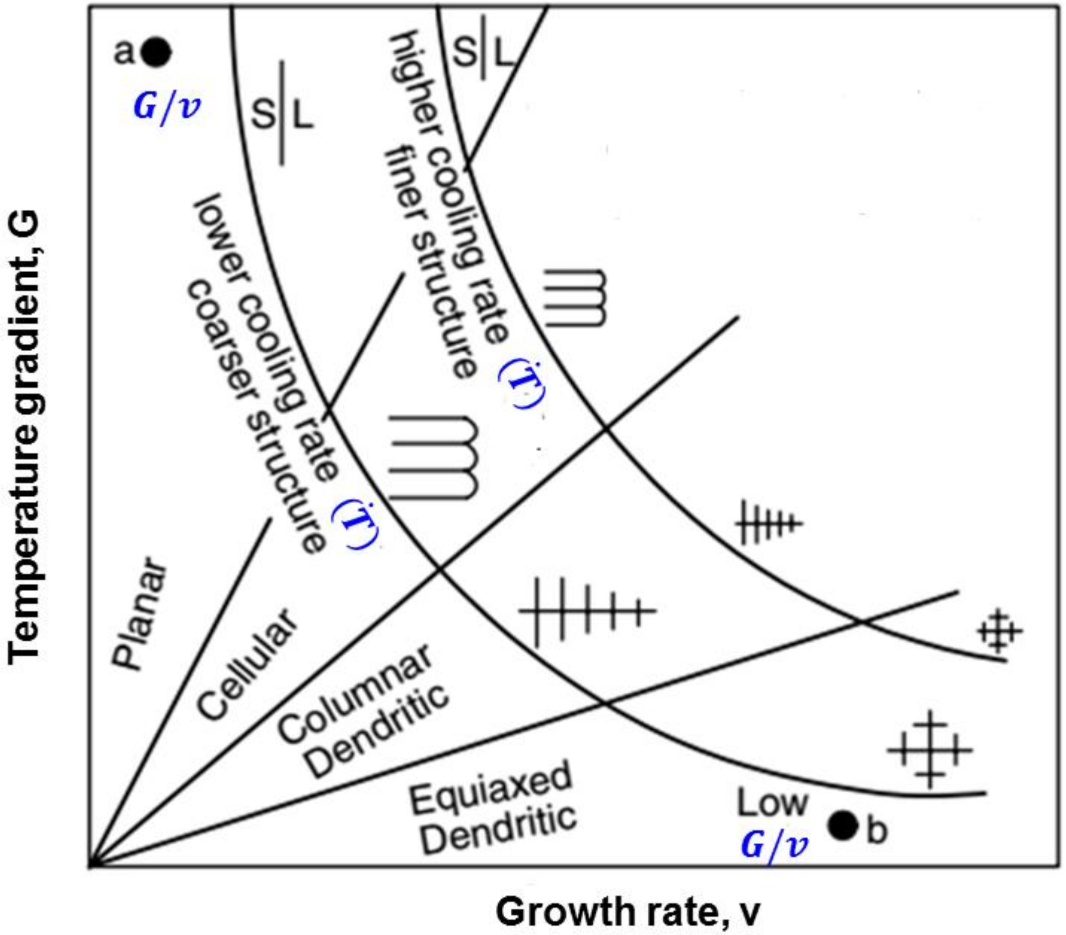
- $G \; X \; v$ determines the size of solidification microstructure

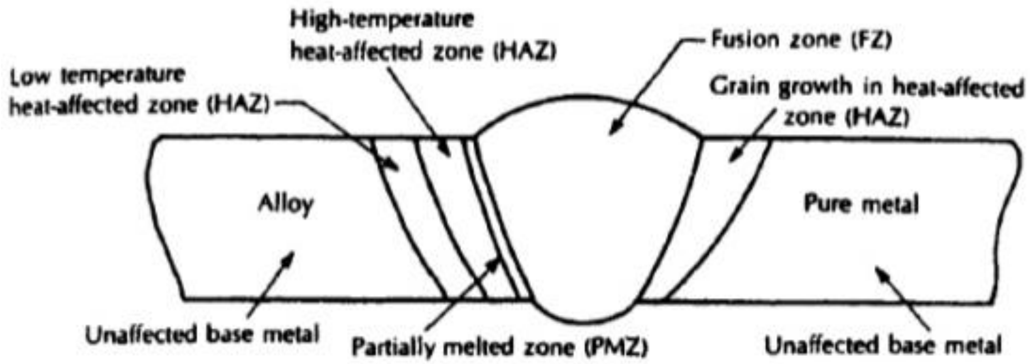
$G = \frac{dT}{dx}$ and $v = \frac{dx}{dt}$

$$G\,.\,v = \frac{dT}{dx}.\frac{dx}{dt} = \frac{dT}{dt} = \dot{T} \quad : Cooling\ rate$$

G/v: *determines morphology of solidification structure*

$\dot{T} = G\ X\ v$: *determines size of solidification structure*





1) Fusion zone 2) Weld interface/partially melted zone
3) Heat affected zone 4) Unaffected base metal

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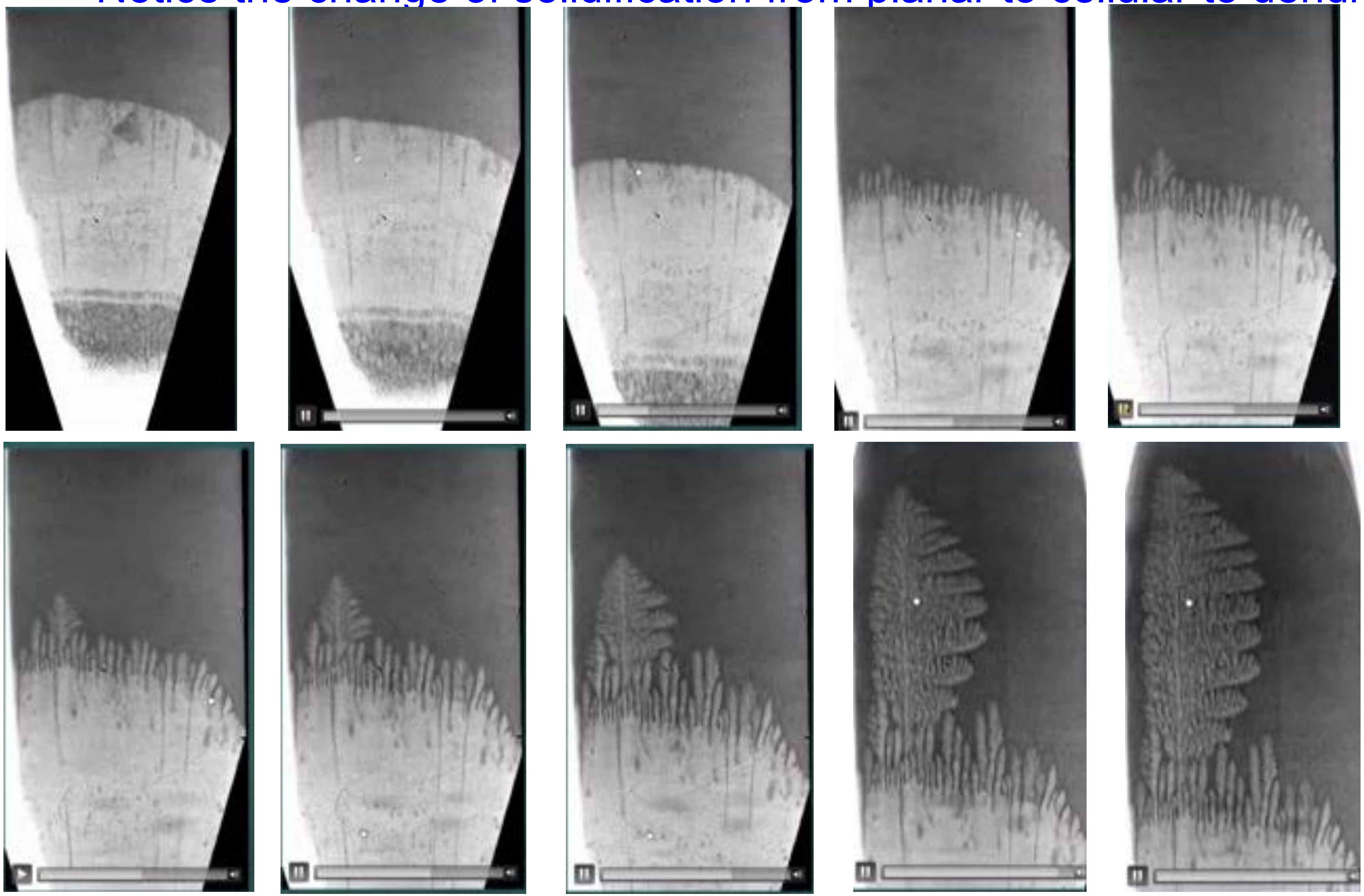
Notice the change of solidification from planar to cellular to dendritic mode



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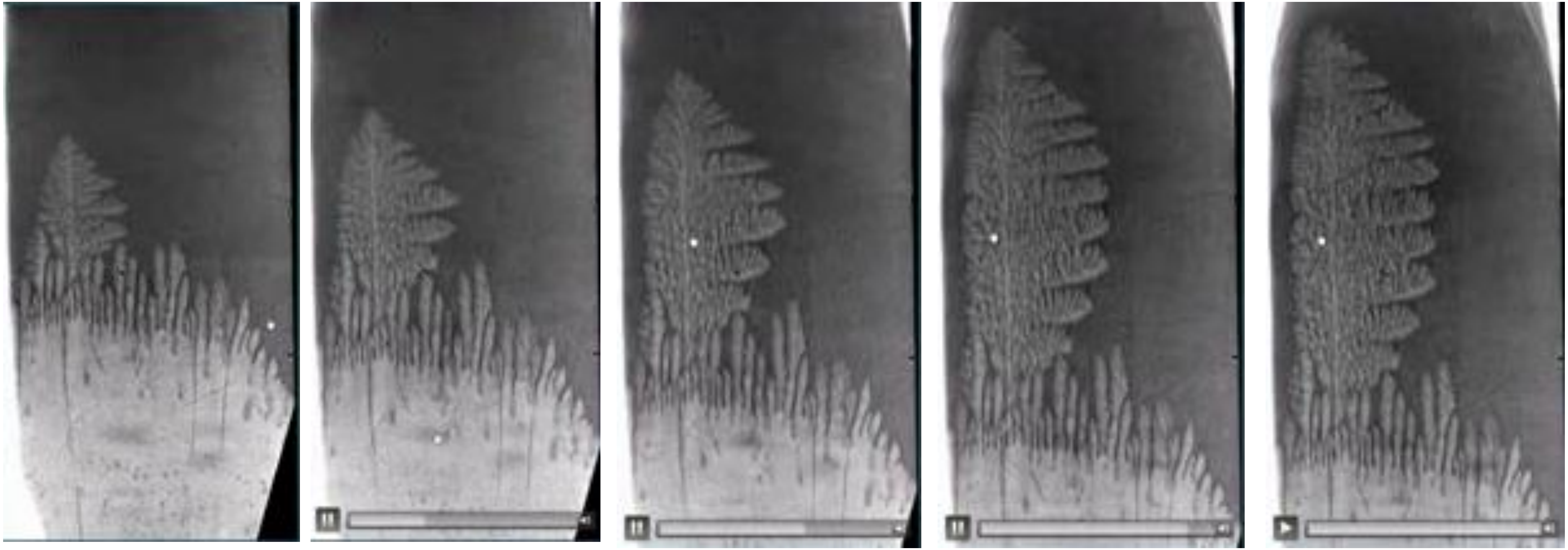
Notice the dendritic mode of growth for long distance. This will lead to columnar grains



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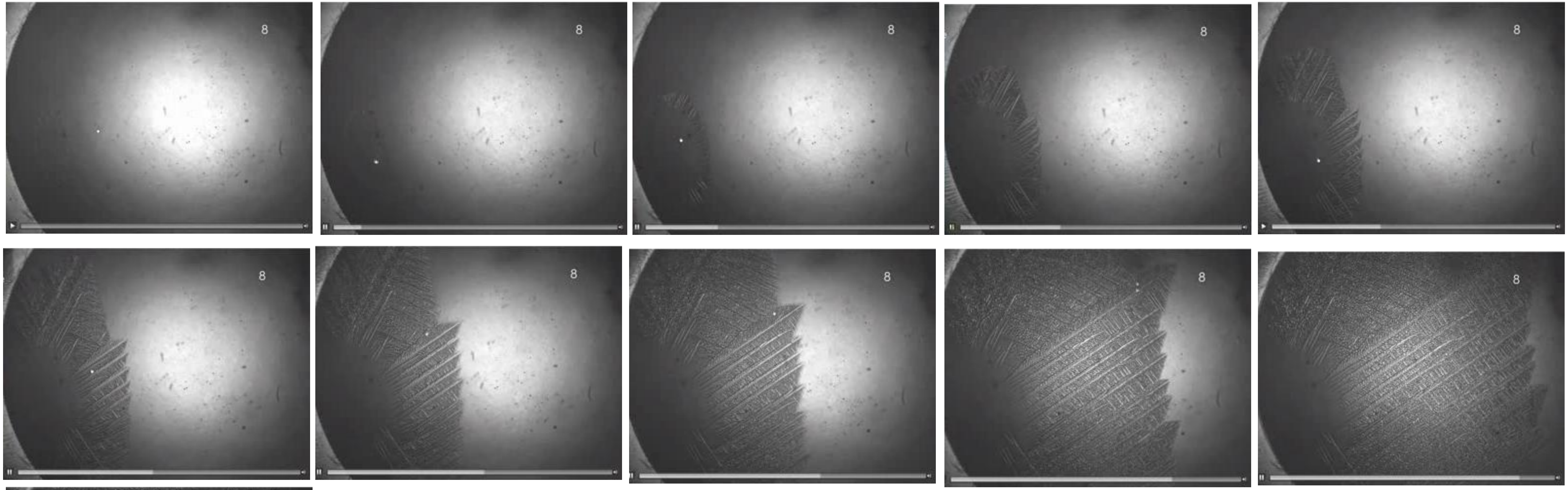
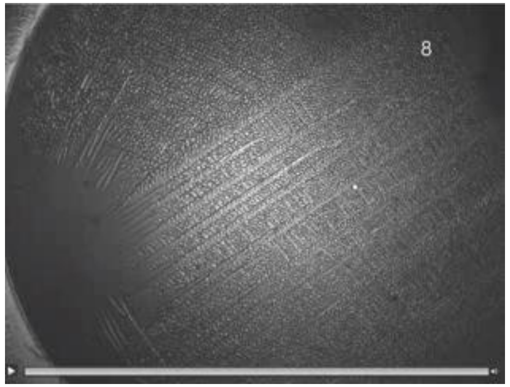
Notice the change of solidification from planar to cellular to dendritic mode.
Long dendrites have channels of liquid in between





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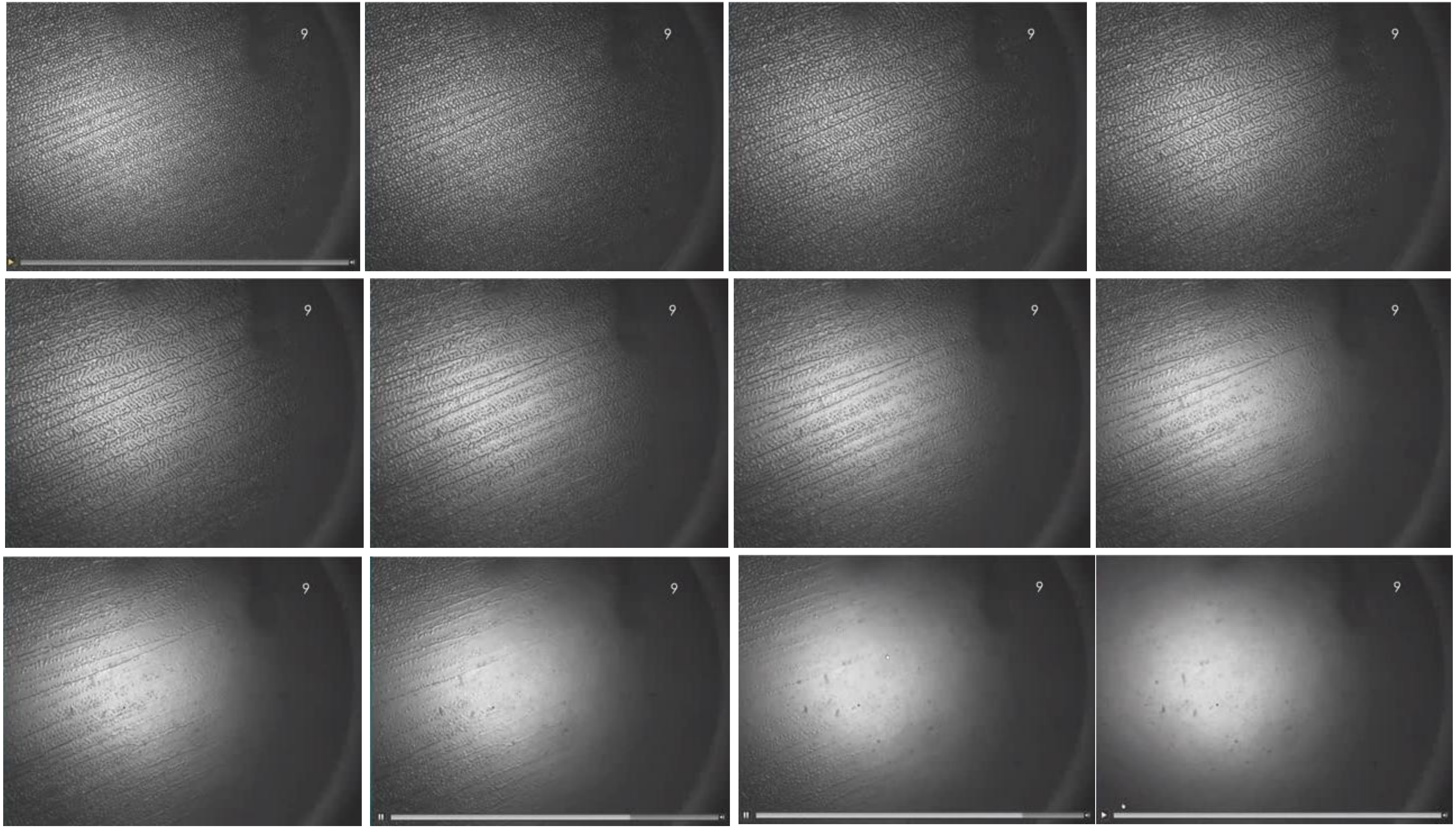
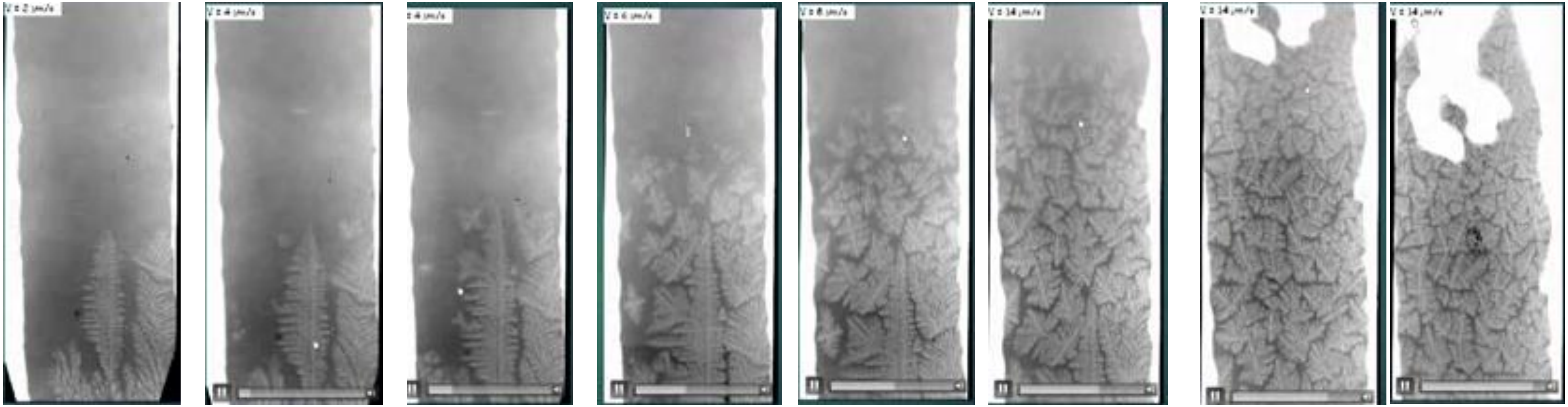
Notice the loss of contrast as the last to solidify liquid disappears.
In a metallic alloy, these channels can be etched out.



Notice the change of solidification from columns of dendrite to equiaxed dendrite.

V = 2 m/s | V = 4 m/s | V = 4 m/s | V = 6 m/s | V = 8 m/s | V = 14 m/s | V = 14 m/s | V = 14 m/s



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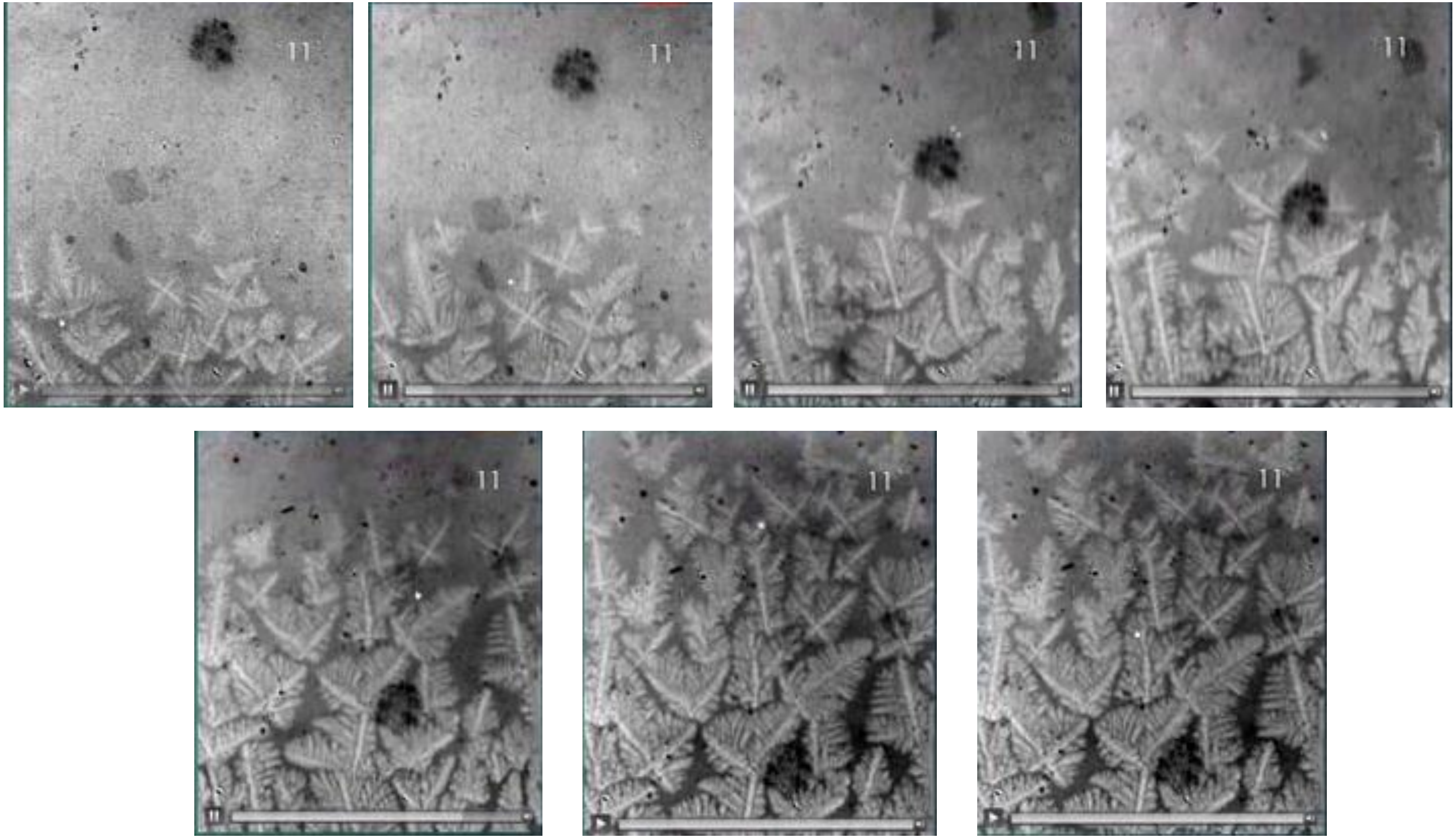
Notice the profuse nucleation of grains that grow as dendrites to form equiaxed microstructure.



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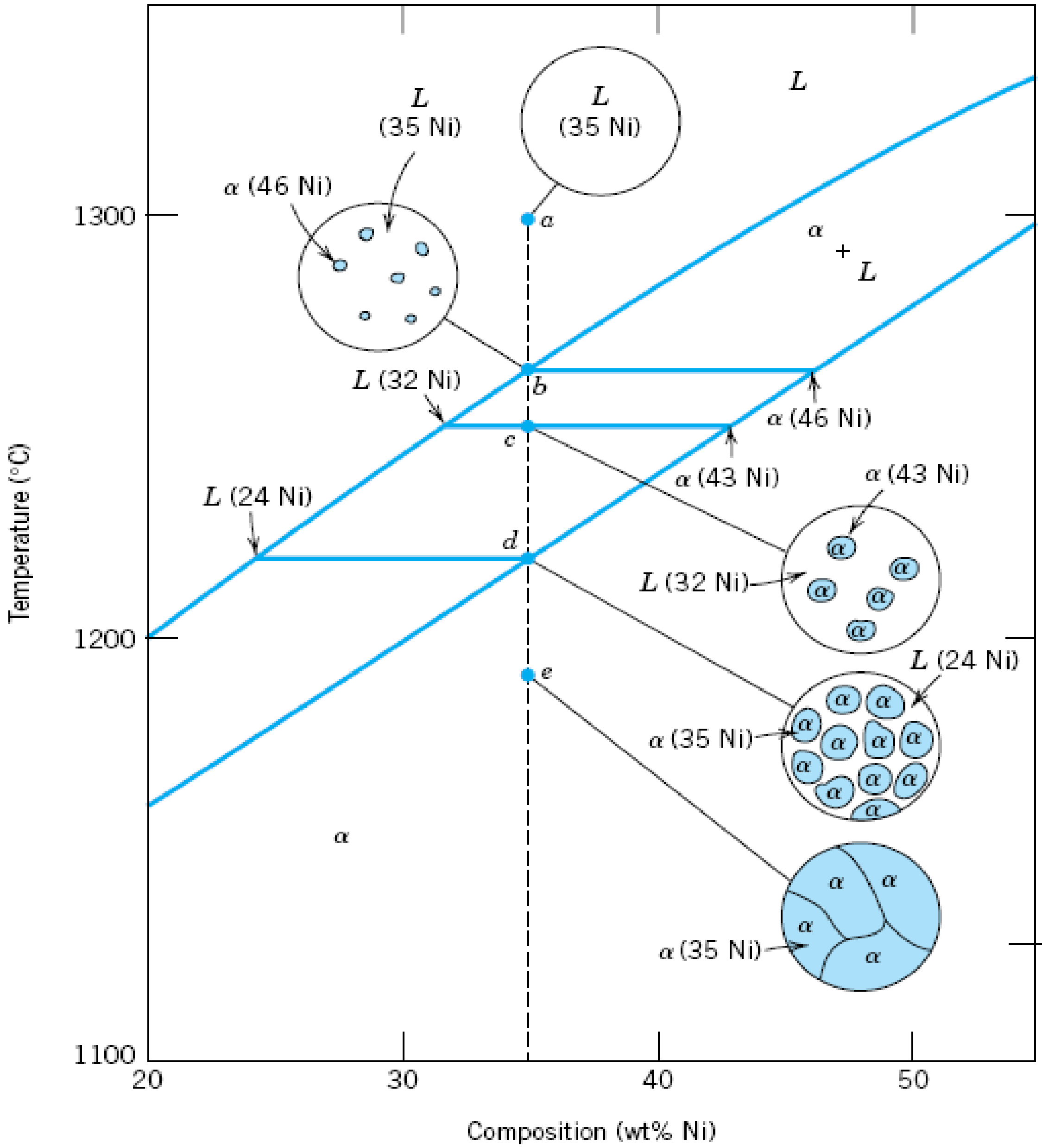
Microstructure evolution during solidification



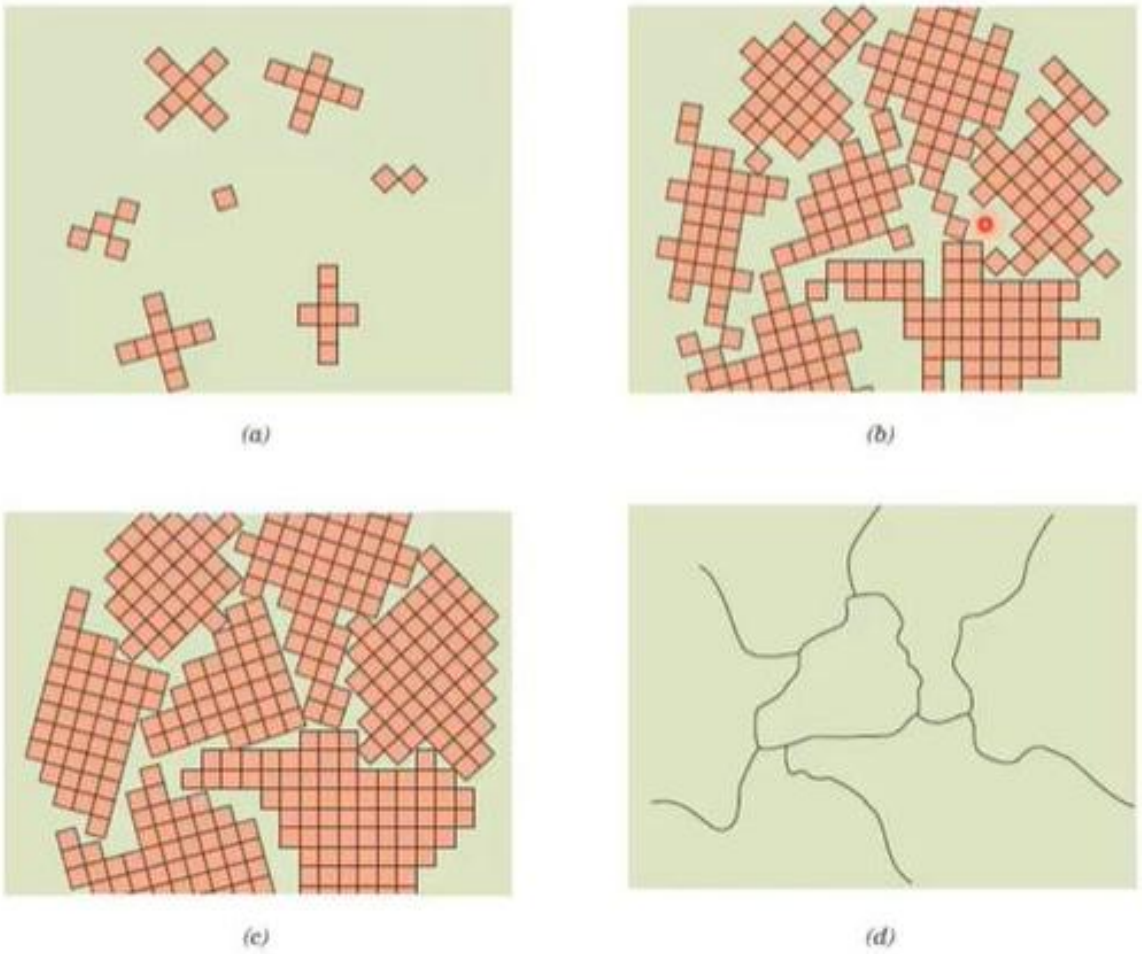
Single phase polycrystalline α

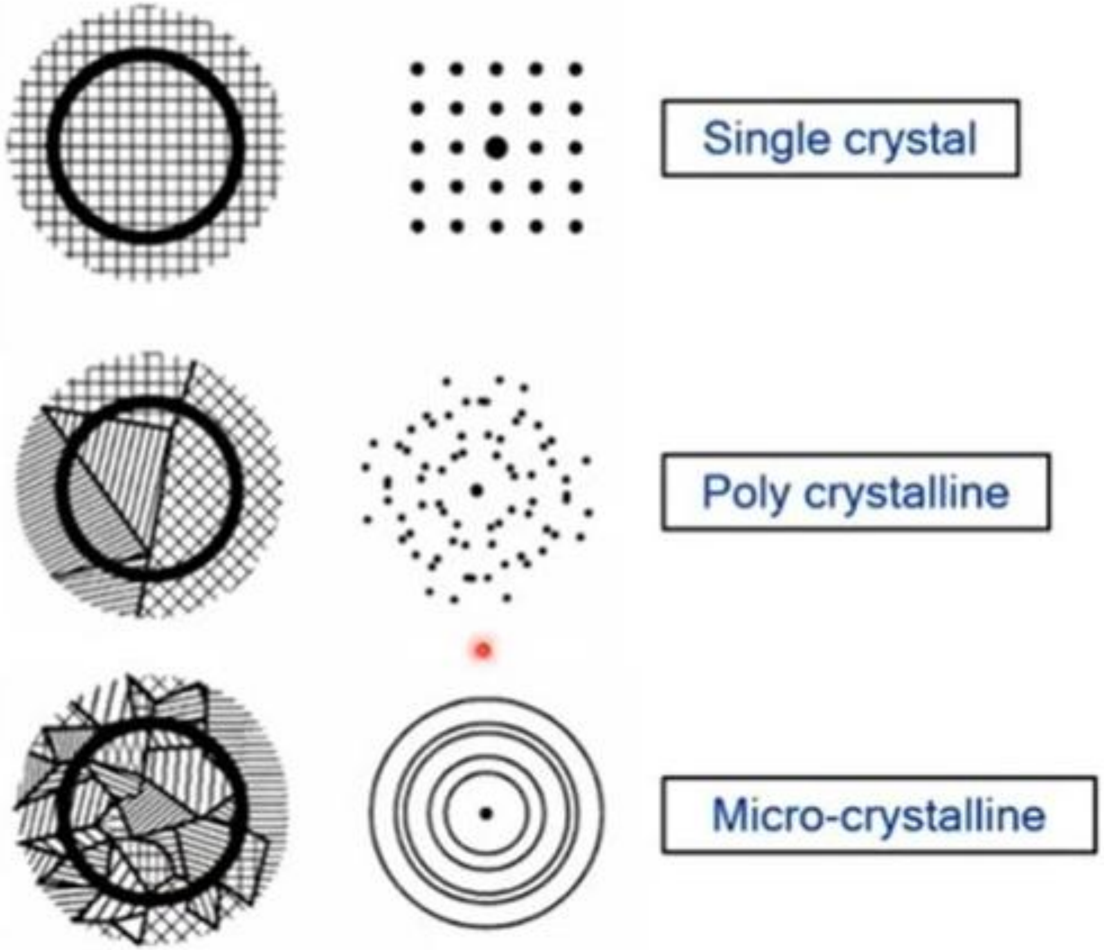
Lever Rule

$$f_\alpha = \frac{35-32}{43-32} = \frac{3}{11} = 0.273$$

$$f_L = 1 - f_\alpha = 0.727$$

Crystalline Materials: Single *Vs.* Polycrystalline



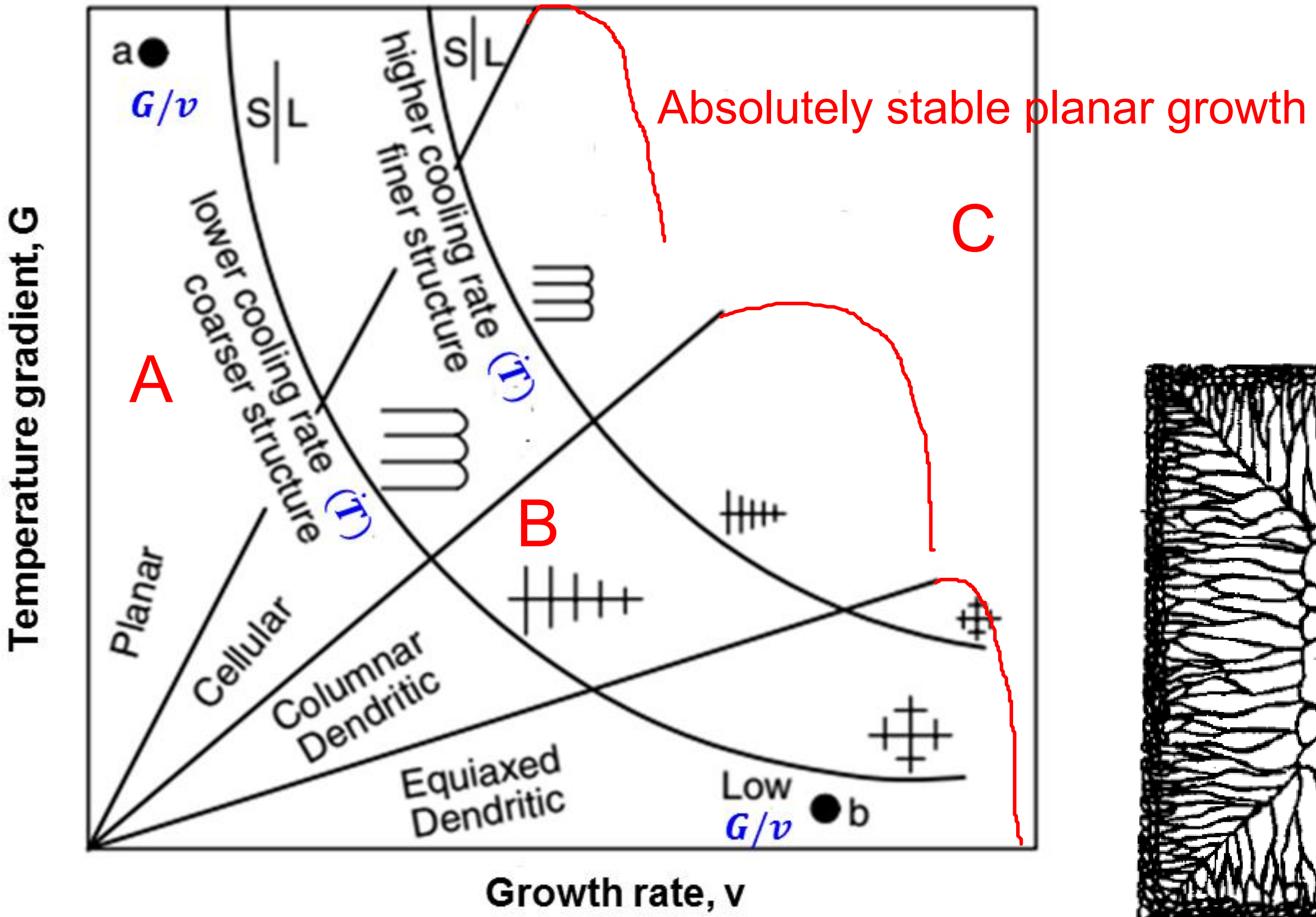


Diffraction Pattern

Complete microstructure map

G/v: *determines morphology of solidification structure*

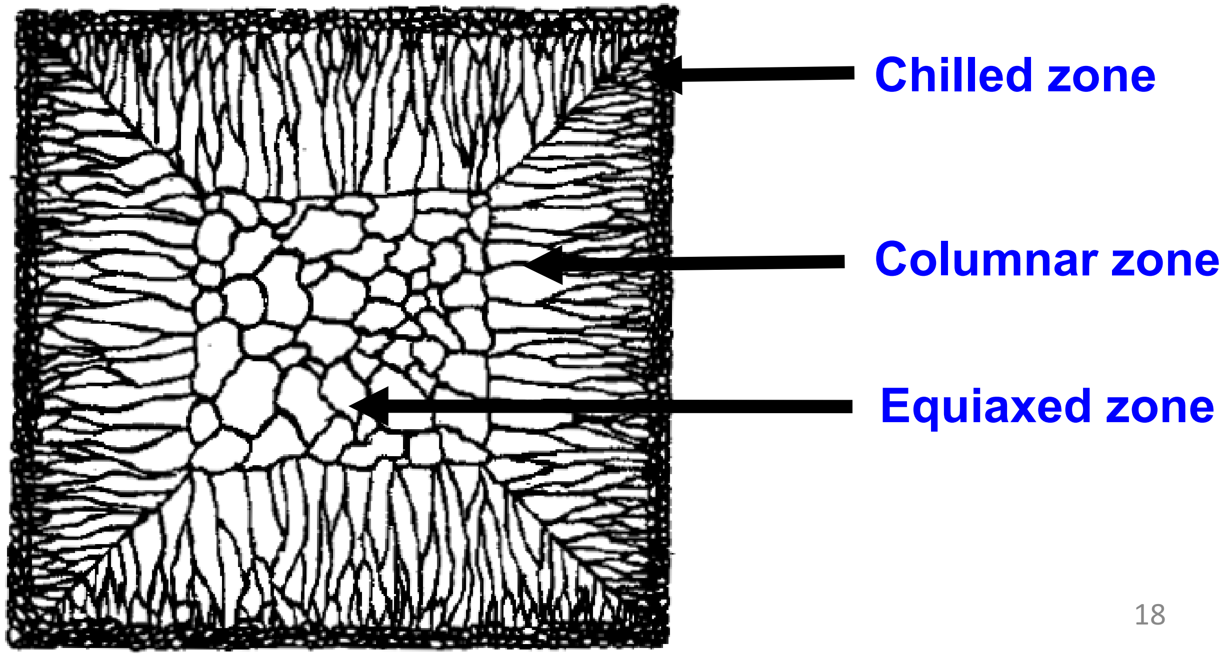
$\dot{T} = G \, X \, v$: *determines size of solidification structure*



A: Single crystal growth

B: Casting process

C: Laser welding process

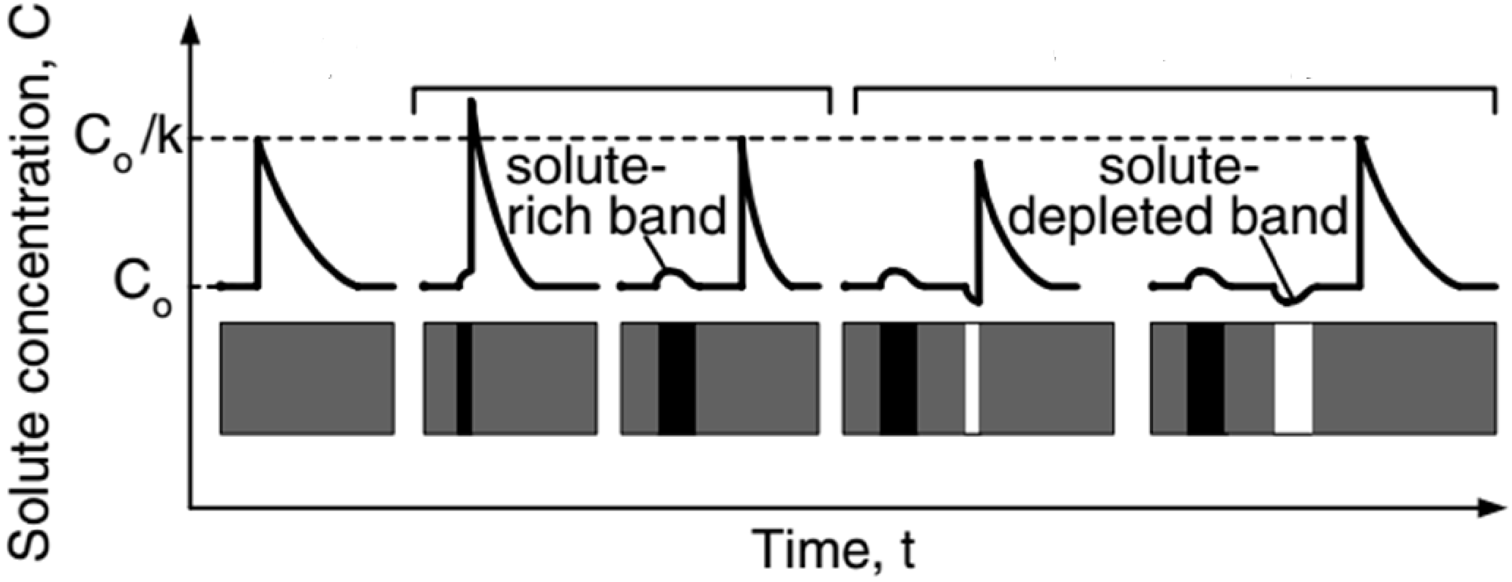


Banding during solidification of fusion welds

- Small changes in growth rate (*v*) leading to formation of bands in the microstructure or change in solute content.

$$\delta = \frac{2D}{v}$$

- Sudden increase in velocity results small increase in composition in boundary layer
- Sudden decrease in velocity results small decrease in composition in boundary layer



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Thank you